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Biodiversity of most dead wood-dependent organisms in thermophilic temperate oak woodlands thrives on diversity of open landscape structures

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ABSTRACT

Oak and mixed deciduous forests with oaks are the most widespread woodland types in the central European lowlands. The aim of this study was to analyse how the biodiversity of saproxylic organisms (fungi, lichens, beetles, and ants, bees and wasps) in thermophilic temperate oak woodlands respond to the openness in landscape structure of tree habitats. We sampled 32 sites in a split-plot design in Krivoklatsko (Czech Republic), which were chosen to include spatial diversity, including dense forests, open forests, woodland edges and solitary trees. A canonical correspondence analyses (CCA) and generalized additive models (GAM) were used for analyses. The results indicated that the taxa studied showed differences in species composition among the studied landscape structures and most taxa preferred more open and light conditions of the woodland environment. We also observed positive effect of the heterogeneity in open landscape structures on biodiversity of saproxylic organisms. As it is recently showed by ecologists, most of the thermophilic oak woodlands are threatened by succession, saproxylic organisms are facing decline throughout the world and traditional forest management (e.g. game keeping, wood pasturing or coppicing) appears to be one solution to mitigate biodiversity loss.

1. Introduction

Many parts of the Earth are facing biodiversity loss and many places important for their biological legacy are negatively affected by the homogenisation and intensification of recent environmental management methods (Vitousek et al., 1997; Dirzo and Raven, 2003). Many species are predicted to be lost before they have been explored for science, and in the best case scenario, specimens are at least currently preserved in several museum collections around the world (Ponder et al., 2001; Horak et al., 2012). The number of threatened species is increasing and these species are losing their former strongholds and survive on the edges of their distribution areas (Channell and Lomolino, 2000). Many formerly common species and their habitats have been vanishing on various landscape scales (Konvicka et al., 2008). However, in contrast, some species are increasing in abundance and distribution (e.g. extralimitally) or are colonising new disjunctive areas (Kloot, 1984). Most probable causes are changes in land use and climate (Horak et al., 2013a).

Woodlands dominated by oaks are recently facing the abandonment of traditional forest management, and the modern forest management of high forests (Peterken, 1993) together with artificial regeneration are probably factors that have jeopardized their maintenance (Mielikainen and Hynynen, 2003). Wide areas of woodlands at lower altitudes have been pastured in the past by domestic animals (cattle, pigs, sheeps and goats), intensively coppiced or coppiced with standards, gathered for litter, used for game keeping and hunting and many disparate management types have created the fine mosaic of woodland patch types in which many species thrive (Vera, 2000; Horak et al., 2012). The abandonment of such management activities and the creation of conservation areas with unmanaged oaks have led to the expansion of early successional tree species such as birch, which shade lower canopy strata and overgrow solitary trees with limbs in the lower canopy (Ranius and Jansson, 2000). A similar problem is observed with maples and ashes on high nutrient soils, beech at high altitudes on mesic (or deep) soils and with conifers (spruce and pine) artificially planted under the oak canopy (Hofmeister et al., 2004; Janik et al., 2011).

The loss of formerly common species is reported also from temperate oak woodland areas (Buse et al., 2007; Hedl et al., 2010). Oak and mixed deciduous forests with oaks are the most

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widespread woodland habitats in the central European lowland and adjacent areas (Svoboda et al., 2011). In the temperate climatic zone, various types of oak woodlands naturally occur up to ca. 600 m a.s.l., especially on dry and semi-dry sites (Neuhauslova, 1998). The majority of European oak species are sun-loving trees, whose recruitment is dependent on favourable micro-climatic conditions during the fruiting stage of mature trees and for seedling growth and development (Rolecek, 2005; Ugurlu et al., 2012). Oaks are also known for their stamina and high quality wood and are often protected as single specimens or monument trees. Furthermore, acorns are nutrient-rich feed for domestic and semi-domestic animals such as pigs or wild boars (Vera, 2000). Veteran oaks are one of the most suitable hosts for organisms that are dependent on dead wood (Svoboda et al., 2011; Alexander, 2013).

Organisms dependent on dead wood, like beetles and fungi, have recently been the focus of forest ecology (Lindhe et al., 2005; Buse et al., 2007; Norden et al., 2008). These organisms are highly interconnected with woodland habitat types and many are bioindicators (Svoboda et al., 2010). Studies performed at the species level have shown that the openness of the surrounding woodland environment often leads to an increase in species population density (Buse et al., 2007). A similar situation is observed in studies at guild or family levels, where most taxa thrive on canopy openness (Lindhe et al., 2005; Horak and Rebl, 2013). Although a gap remains in the knowledge of multi-taxon responses, and non-boreal landscapes are under-represented with respect to the number of studies. Namely, species diversity is much higher in temperate and Mediterranean woodlands, and oak-dominated or mixed woodlands are European hot-spots of biodiversity (Blasi et al., 2010; Buse et al., 2010). Solitary oak trees are known to be very rich with respect to the number of species of disparate organisms (Svoboda et al., 2010). Nevertheless, the responses of dead wood-dependent organisms to environmental changes can be of high interest for suitable biodiversity management activities.

The aim of this study was to analyse how the biodiversity of saproxylic organisms responds to the openness and diversity in landscape structure of tree habitats in thermophilic temperate oak woodlands, using a multi-taxa approach.

Namely, we focused on (i) species composition, (ii) richness and (iii) its trends of four taxa (fungi, lichens, beetles, and ants, bees and wasps) and their influencing by four landscape structures (solitary trees, woodland edges, open forests and dense forests).

2. Methods

2.1. Study area

Krivoklatsko is situated in the western part of the Central Bohemian range about 20 km west of Prague (coordinates: 50.00N; 13.88E). It is known within the Czech Republic for a high biodiversity, which has recently been highlighted by attempts to protect this area as a national park (Hula, 2009). Its large area (ca. 630 km^2) of low and medium altitudes (ca. 400 m a.s.l.) contains diversified land use, geomorphology and a large number of disparate nature conservation and cultural heritage areas, such as Krivoklatsko UNESCO Biosphere Reserve and Krivoklatsko Protected Landscape Area, with nearly 30 small-area protected sites. Krivoklatsko is one of the most forested parts of the Czech Republic, with 62% covering by forests (Hula, 2009).

2.2. Study taxa

We studied four taxa: fungi (Fungi), lichens (Lichenes), beetles (Coleoptera) and ants, bees and wasps (Hymenoptera: Aculeata). Only species with an affinity for dead wood habitats (hereafter called saproxylics) were evaluated. For beetle species, all obligate saproxylics were taken into account (Schmidl and Bußler, 2004). All dead wood-dependent fungal species (Hagara et al., 2005) were analysed, except those with corticioid fruiting bodies; pyrenomycetes and inoperculate discomycetes with small apothecia were also not sampled. Most saproxylic species of hymenopterans use dead wood for nesting and some are able to build nests in other substrates, thus, we only evaluated species with a predominant dead wood strategy (Macek et al., 2010). The associations of lichens with dead wood substrates in central Europe are more complicated than for the other studied taxa (Jansova and Soldan, 2006). As lichens are mostly long-lived organisms, it is often hard to determine whether a species is associated only with a living tree (i.e. arboricolous) or is more opportunistic and dependent only on substrates generated by wood plants. Therefore, we used all lichen species in our analyses that were found on dead and partly living wood, because most trees in old-growth forests include many dead wood microhabitats (Winter and Moller, 2008).

2.3. Sampling of taxa

Fungi and lichens were sampled using circular plots with a radius of 20 m surrounding the traps (area 1256 m^2). For solitary trees, the shape of the sampling plot was adapted to the tree-dominated habitat in the surroundings, but also had a fixed total area. The centre of the sampling plot was 20 m from the edge of the woodland for the edge category. Fungi were sampled four times in June, September and October 2011 and May 2012. Lichens were sampled once during August 2011. Beetles and hymenopterans were sampled using large window trunk tree traps (Horak, 2011), which are known to be highly effective for trapping dead wood associates including flightless fauna (Horak et al., 2013b). The traps were active and regularly emptied during the 2011 vegetation season (May–September).

2.4. Study environmental variables and their sampling

Our study focused on four main landscape structures of tree habitats:

(i) solitary trees (canopy openness: mean = 32.51% ± 4.57 SE) with mean distance of 94.75 m (SE = 22.22 m) from the nearest forest edge and high coverage of plants and shrubs in the understorey,
(ii) natural woodland edges (32.05% ± 5.92 SE) between forest and non-forest area (grasslands and arable land) with medium coverage of plants and shrubs in the understorey,
(iii) forests with open canopy conditions (32.87% ± 4.05 SE) and medium coverage of plants and low coverage of shrubs in the understorey,
(iv) dense forests with closed canopy conditions (12.88% ± 0.68 SE) with low coverage of plants and absence of shrubs in the understorey.

First three landscape structures (i–iii) reflected different open canopy conditions and one (iv) reflected closed canopy conditions.

In total, we sampled 32 sites that were designed as split plots – i.e. eight whole plots contained all four landscape structures. Three pairs of whole plots were distributed in or near three protected areas recently listed (www.pralesy.cz) as the most natural woodlands in the Czech Republic (Brdatka, Nezabudicke skaly and Na Babe) and two pairs between protected areas Jouglovka and Kohoutov. The main reason for this choice was natural or semi-natural tree species composition of above mentioned protected areas and their surroundings (Nemec and Lozek, 1996).

Coordinates (latitude and longitude) of the plot centres as the spatial terms were measured directly in the field using a Garmin Colorado 300.

The amount of dead wood as the best reflections of habitat area in woodland habitats for saproxylic organisms was measured in each plot. All dead wood (mean = 1.20; SE = 0.26; min = 0.01; max = 4.60 m^3) was measured, including all lying and standing pieces to 5 m above the ground and with a diameter >0.10 m within a 20 m radius surrounding the traps. When only thinner diameter dead wood was found, we used a value of 0.01 m^3.

Canopy openness (mean = 27.58; SE = 2.54; min = 10.36; max = 61.09%) was measured using a Nikon COOLPIX 995 camera with a Nikon FC-E8 Fish Eye converter.

2.5. Data analyses

All analyses were performed in CANOCO. We used a presence/absence species matrix as it gave a better final comparison of taxa sampled using different techniques (Horak et al., 2013c). A canonical correspondence analysis (CCA) was used as the most appropriate method of multivariate statistics. Landscape structures were used as categorical variables and dead wood with coordinates and their crossed and quadratic products were treated as continuous co-variables (Horak, 2013). A global permutation test was set at 9,999 permutations under the full model. The permutation type was restricted for a split-plot design with the whole plot freely exchangeable. Due to the presence of empty species samples for fungi, lichens and Hymenoptera, we used the virtual opportunistic species present in all samples. The potential bias of multi-colinearity was detected using the variance inflation factor (VIF). The joint effect of the studied categories of landscape structures with co-variables was computed using the process of variance partitioning. The Trace/Total interia ratio was used for comparison of the CCA among the studied taxa. Generalized additive models (GAM) were used to compute and visualise species richness gradients for each studied taxa in CanoDraw. The Akaike information criterion (AIC) was used to select the best response of the species richness (number of species as a dependent variable) to the studied categories. All photographs of canopy were evaluated using a Gap Light Analyser 2.0. The difference in canopy openness between dense and open forests was assessed by paired t test.

3. Results

We observed 78 species of saproxylic fungi, 36 species of lichens, 153 species of beetles and 32 species of hymenopterans (Tables S1–S4).

The control analysis showed that there was a significant difference in canopy openness between dense and open forests ($t = -5.21$; $P < 0.01$).

The effect of the studied landscape structures (solitary trees, woodland edges, open forests and dense forests) on species composition of all studied taxa was significant and independent contribution was nearly the same for all taxa. Lichens showed the highest independent and joint contributions and beetles shared the lowest contribution joint with co-variables (Fig. 1).

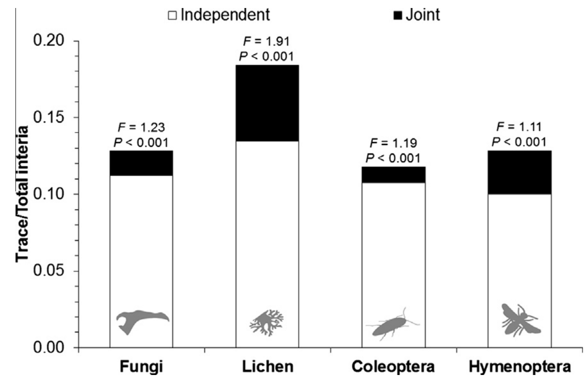

Fig. 1. Comparison of results of canonical correspondence analyses (CCA) on species composition of the studied taxa (fungi, lichen, Coleoptera, and Hymenoptera) in the thermophilic oak woodland area. Independent contribution of the studied categories (solitary trees, woodland edges, open forests and dense forests) is white and joint contribution with space and dead wood as co-variables is black. F and P values of CCA are indicated above joint contributions.

Most of the studied taxa showed a distinct species composition among the studied landscape structures (e.g. fungi in dense forests and edges; Fig. 2a). The results also showed that the species composition of most taxa was almost disparate in dense forests, compared to other habitats (Fig. 2a–c), and were also poorer in number of species in most cases (Fig. 2). Fungal communities were most similar in open forests and solitary trees (Fig. 2a). The species richness of lichens appeared to increase with an increasing gradient of openness in landscape structures (Fig. 2b). Beetles preferred solitary trees (Fig. 2c), whereas hymenopterans showed the highest preference for ecotonal habitats and the species composition here was most distinct from that of other habitats (Fig. 2d).

The species richness isolines did not illustrate any clear trend with respect to the studied design for fungi, although the samples from the edges and dense forest habitats showed a bit higher species richness than other habitats (Fig. 3a), which is in agreement with previous results (Fig. 2a). The species richness of lichens and beetles decreased from solitary trees towards closed forests (Fig. 3b and c). Open forests and edges possessed the highest species richness of saproxylic Hymenoptera (Fig. 3d).

4. Discussion

The results showed that most taxa preferred diversification of the more open and light conditions of the woodland environment. Highly diversified open canopy structures like solitary trees, natural edges and open forests are potentially primary objectives of the present forest management (Salek et al., 2013). Furthermore, some saproxylics, namely beetles and lichens, preferred solitary trees, which are usually not considered as a forest habitat. This might lead to the conclusion that some saproxylic organisms do not prefer forest. Although, the response of the studied taxa might be influenced by the fact that open thermophilic oak woodlands in our study area have a high temporal continuity (Neuhauslova, 1998). Oak woodlands have a naturally higher canopy openness (Rolecek, 2005; Ugurlu et al., 2012), due to the requirements of the dominating tree species (*Quercus* spp.) – associated organisms are thus more adapted to light conditions in woodlands.

Except for the natural fauna and flora, open canopies promote the presence of species adapted to warm climatic conditions (Ugurlu et al., 2012; Alexander, 2013). This is well-illustrated by the occasional occurrence of the Mediterranean long-horned beetle, *Purpuricenus kaehleri* (L.) in the Czech Republic. This species is not threatened in Europe (Nieto and Alexander, 2010) and in the Czech Republic, is mainly distributed in the two most thermophilic areas: Krivoklatsko and southern Moravia (Slama, 1998). The species was considered in the Red-list as regionally extinct (Farkac et al., 2005), but was once again collected in our study area in present years, probably due to favourable climate conditions.

Oak woodlands have often been managed in forest systems known for higher canopy openness, such as coppices, coppices with standards or pasture woodlands (Vera, 2000; Horak et al., 2012; Fartmann et al., 2013). Currently, there are only a few over-matured stands in the Czech Republic, which are indicative

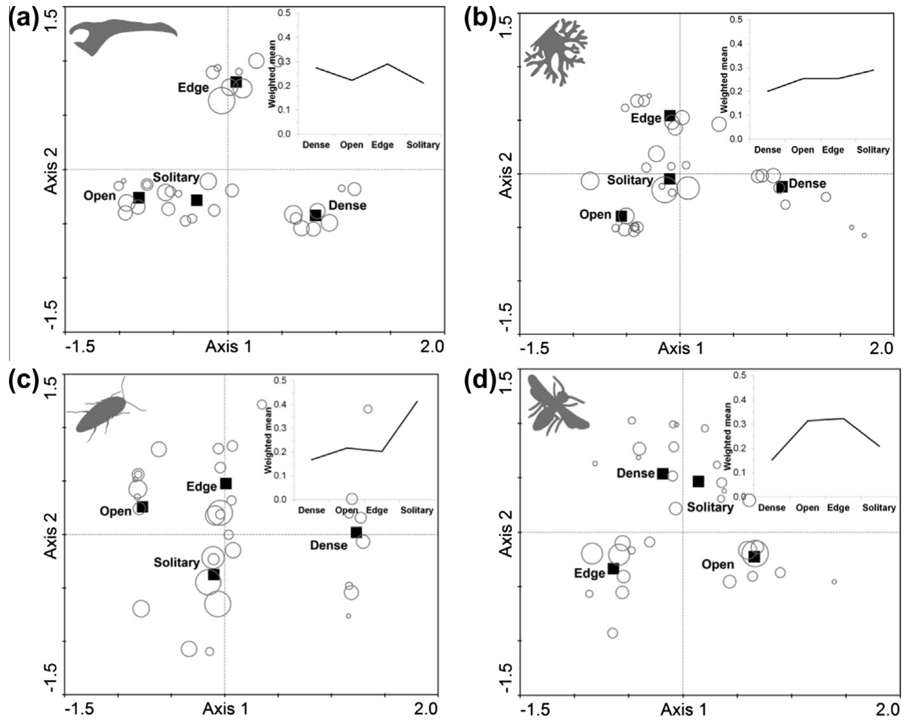

Fig. 2. Sample-environmental attribute plots (a high circle radius shows a high species richness of samples) for the studied taxa: (a) fungi, (b) lichens, (c) Coleoptera and (d) Hymenoptera in the thermophilic temperate oak woodland area. Sub-graphs in the upper right corners show the value of the weighted species mean in each studied category. First and second CCA axes are shown.

of former coppice management and the only recently active coppice-like stands are riparian corridors dominated by alders (Pecha, 2010). Except for presence of coppicing and pasturing, one of the important factors for high biodiversity might be the absence of modern forest management (with the artificial regeneration of stands after harvesting), which causes a dominance of oak in vegetative and natural regeneration and also leads to a sparser canopy caused by damage by high game stocks on natural regeneration. In our study area, historical sources also mention a high diversity of other, so-called traditional forest management activities – such as litter gathering or basketry, which also promote an open forest structure (Horak et al., 2012).

The high area of the Krivoklatsko woodlands was managed in the past for game hunting, which has been carried out since medieval times – i.e. a vast area was fenced in as a game park for the Czech king and nobility (Horak and Rebl, 2013). The recent hunting community has been criticized for high game stocks, which cause damage to natural regeneration and timber production (Kupcak, 2005). However, it is known that a high game stock pressure is a successful biological agent that acts against forest succession and towards closed canopy woodlands dominated by beech and conifers (Janik et al., 2011).

The main problem of closing canopies and following biodiversity loss, is that nearly all traditional forest management types are restricted (e.g. wood pasturing) or are affected by Czech law (e.g. coppicing; Utinek, 2004). Except for commercial forestry, the only recent effect of conservation management is to leave the forests unmanaged (Hort et al., 1999) – however, this conservation activity mostly leads to canopy closure (Horak et al., 2012) and to the dominance of late successional tree species such as beech, especially in thermophilic oak forests on deep soils. Non-intervention management in thermophilic oak woodlands mostly leaves open structures only in stands on slopes, with shallow rocky soils, a lower nutrition quality and a humus layer caused by natural processes such as erosion (Rolecek, 2005) and by a higher accumulation of game due to less activity from hunters.

4.1. Management possibilities

Thermophilic oak forests on deep soils represent successionally unstable vegetation created by man (Svoboda, 1943; Rolecek, 2005). Thus, their survival and spatial structure is dependent on active forest management. Traditional management activities, such as coppicing or wood pasturing, represent one solution (Hedl et al., 2010; Hartel et al., 2013) – however, these have been highly problematic in recent times, as mentioned above.

Game parks, such as Lany Game Park within the studied area, with its large area of 30 km^2, represent one management possibility towards open canopy woodlands (Horak and Rebl, 2013). There are also many ways to manage commercial stands with high forest, which could (at least ephemerally) mimic the openness of forest management – these are larger areas with prolonged shelter-wood

Table 1
Species richness trends in the thermophilic oak woodland. Results of GAM using selection based on AIC, significant *P* values are in bold.

Taxa	First CCA axis DF	Second CCA axis DF	Resid. DF	*F*	*P*	AIC
Fungi	1	6	24	4.54	**<0.01**	299.16
Lichen	1	4	26	2.61	**<0.05**	263.21
Coleoptera	4	4	23	2.78	**<0.05**	2472.58
Hymenoptera	1	6	24	3.44	**<0.05**	93.63

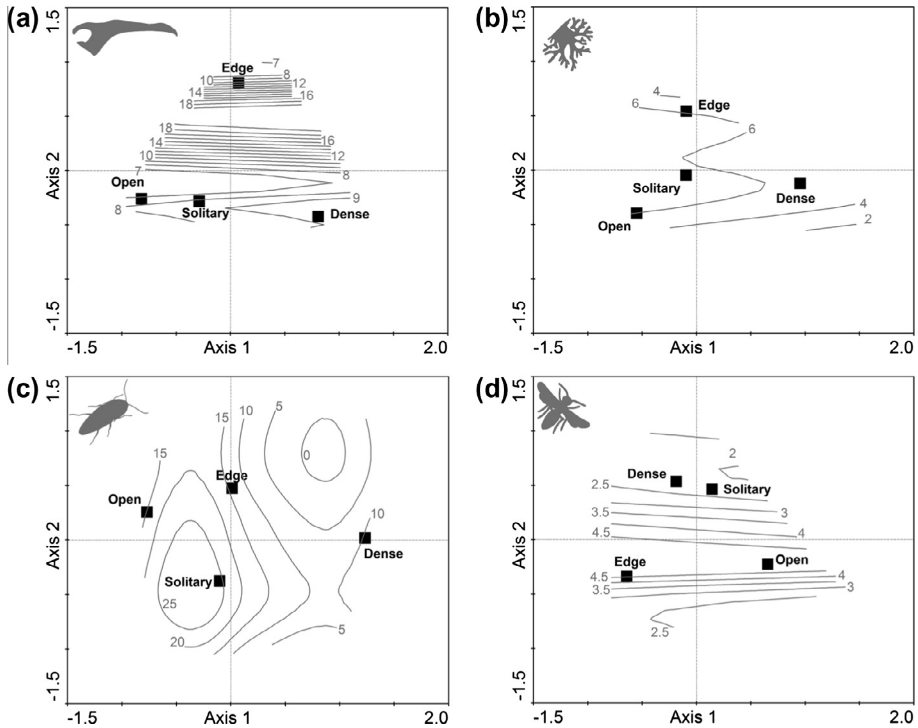

Fig. 3. Response of particular taxa: (a) fungi, (b) lichens, (c) Coleoptera and (d) Hymenoptera to the studied categories (black) in the thermophilic temperate oak woodland area. Grey isolines reflect the trend (not total value) in species richness derived from GAM (Table 1); the first and second CCA axes are shown.

cuttings, leaving several trees (more damaged = more effective) in clear-cuts, or the maintenance, promotion and planting of solitary individuals or avenues in grasslands, arable land and along roads.

The worst solution for the protection of regional biodiversity of thermophilic oak woodlands appears to be to leave the stands unmanaged. This solution, if successfully practiced in beech woodlands (Gossner et al., 2013), is applicable only to dry forests on shallow rocky soils, which are naturally disturbed by soil erosion, or semi-naturally by game stocks.

5. Conclusions

The first topic that needs to be addressed is the restoration of traditional forest management systems such as coppicing and wood-pasturing, at least in conservation areas. This is because these management types retain the open structures and avoid the dominance of late successional tree species, namely beech, which has lower commercial potential than oak and furthermore, the low commercial potential of coppice-like management appears to be a pseudo-issue in recent times (Utinek, 2004; Kupcak, 2010).

The second goal is the maintenance of high game stocks, even when they are distributed outside the game parks. Whereas wood pasturing is prohibited within the Czech Republic, the only way to mimic this formerly widely used management method is via the maintenance of high game stocks, especially in conservation areas, where their damage has a low commercial effect. Furthermore, game hunting is known to increase local incomes and improve human diet (Hawkes et al., 1991; Wilkie and Carpenter, 1999).

The third aim is support for local activities towards the maintenance and restoration of solitary trees in non-woodland landscapes, an activity that is often performed by non-governmental organisations.

Semi-natural oak woodlands are one of the most widespread forest types in the lowlands of Central Europe (Svoboda et al., 2011), although since they represent successionally unstable vegetation created by man (Lozek, 1973), without active forest management, they often succumb to the expansion of late successional trees (Rolecek, 2005). The biodiversity of the studied taxa, in most cases, is driven by openness and diversity in the landscape structures of thermophilic oak woodlands. There are many rare or threatened saproxylic organisms connected with oak tree habitats (Buse et al., 2007; Norden et al., 2008; Svoboda et al., 2011; Petrakova and Schlaghamersky, 2011). Saproxylic organisms are in decline throughout the Europe (Nieto and Alexander, 2010) and thermophilic oak woodlands, as one of their main habitats, are threatened by succession (Moga et al., 2009). Thus, active forest management that causes openness and diversification in the management of oak woodlands is one of the solutions to mitigate biodiversity loss.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.foreco.2013.12.018.

References

Alexander, K., 2013. Ancient trees, grazing landscapes and the conservation of deadwood and wood decay invertebrates. In: Rotherham, I.D. (Ed.), Trees, Forested Landscapes and Grazing Animals, pp. 330–337.

Blasi, C., Marchetti, M., Chiavetta, U., Aleffi, M., Audisio, P., Azzella, M.M., Burrascano, S., 2010. Multi-taxon and forest structure sampling for identification of indicators and monitoring of old-growth forest. Plant Biosyst. 144, 160–170.

Buse, J., Levanony, T., Timm, A., Dayan, T., Assmann, T., 2010. Saproxylic beetle assemblages in the Mediterranean region: impact of forest management on richness and structure. For. Ecol. Manage. 259, 1376–1384.
Buse, J., Schröder, B., Assmann, T., 2007. Modelling habitat and spatial distribution of an endangered longhorn beetle – a case study for saproxylic insect conservation. Biol. Conserv. 137, 372–381.
Channell, R., Lomolino, M.V., 2000. Dynamic biogeography and conservation of endangered species. Nature 403, 84–86.
Dirzo, R., Raven, P.H., 2003. Global state of biodiversity and loss. Annu. Rev. Env. Resour. 28, 137–167.
Farkac, J., Kral, D., Skorpik, M., 2005. Červený seznam ohrožených druhů České republiky. Bezobratlí. Agentura ochrany přírody a krajiny. Praha, Czech Republic.
Fartmann, T., Müller, C., Poniatowski, D., 2013. Effects of coppicing on butterfly communities of woodlands. Biol. Conserv. 159, 396–404.
Gossner, M.M., Lachat, T., Brunet, J., Isacsson, G., Bouget, C., Brustel, H., Brandl, R., Weisser, W.W., Müller, J., 2013. Current near-to-nature forest management effects on functional trait composition of saproxylic beetles in beech forests. Conserv. Biol. 27, 605–614.
Hagara, L., Antonín, V., Baier, J., 2005. Velký atlas hub. Ottovo nakladatelství, Czech Republic.
Hartel, T., Dorresteijn, I., Klein, C., Mathe, O., Moga, C.I., Ollerer, K., Fischer, J., 2013. Wood-pastures in a traditional rural region of Eastern Europe: characteristics, management and status. Biol. Conserv. 166, 267–275.
Hawkes, K., O'Connell, J.F., Jones, N.G., Oftedal, O.T., Blumenschine, R.J., Hawkes, K., Blumenschine, R.J., 1991. Hunting income patterns among the Hadza: big game, common goods, foraging goals and the evolution of the human diet. Philos. Trans. R. Soc. Lond. B: Biol. Sci. 334, 243–251.
Hedl, R., Kopecky, M., Komarek, J., 2010. Half a century of succession in a temperate oakwood: from species-rich community to mesic forest. Divers. Distrib. 16, 267–276.
Hofmeister, J., Mihaljevič, M., Hošek, J., 2004. The spread of ash (*Fraxinus excelsior*) in some European oak forests: an effect of nitrogen deposition or successional change? For. Ecol. Manage. 203, 35–47.
Horak, J., 2011. Response of saproxylic beetles to tree species composition in a secondary urban forest area. Urb. For. Urb. Gree. 10, 213–222.
Horak, J., 2013. Effect of site level environmental variables, spatial autocorrelation and sampling intensity on arthropod communities in an ancient temperate lowland woodland area. PLoS One 8, e81541.
Horak, J., Rebl, K., 2013. The species richness of click beetles in ancient pasture woodland benefits from a high level of sun exposure. J. Insect Conserv. 17, 307–318.
Horak, J., Chobot, K., Horakova, J., 2012. Hanging on by the tips of the tarsi: a review of the plight of the critically endangered saproxylic beetle in European forests. J. Nat. Conserv. 20, 101–108.
Horak, J., Hui, C., Roura-Pascual, N., Romportl, D., 2013a. Changing roles of propagule, climate and land use during extralimital colonization of a rose chafer beetle. Naturwissenschaften 100, 327–336.
Horak, J., Vodka, S., Pavlicek, J., Boza, P., 2013b. Unexpected visitors: flightless beetles in window traps. J. Insect Conserv. 17, 441–449.
Horak, J., Peltanova, A., Podavkova, A., Safarova, L., Bogusch, P., Romportl, D., Zasadil, P., 2013c. Biodiversity responses to land use in traditional fruit orchards of a rural agricultural landscape. Agric. Ecosyst. Environ. 178, 71–77.
Hort, L., Tesar, V., Vrska, T., 1999. Forest reserve research network. In: Diaci, J. (Ed.), Virgin Forests and Forest Reserves in Central and Eastern European Countries, pp. 25–44.
Hula, P., 2009. Chráněná krajinná oblast Křivoklátsko. Ochr. Přír. 1, 2–5.
Janik, D., Adam, D., Vrska, T., Hort, L., Unar, P., Kral, K., Horal, D., 2011. Field maple and hornbeam populations along a 4-m elevation gradient in an alluvial forest. Eur. J. For. Res. 130, 197–208.
Jansova, I., Soldan, Z., 2006. The habitat factors that affect the composition of bryophyte and lichen communities on fallen logs. Preslia 78, 67–86.
Kloot, P.M., 1984. The introduced elements of the flora of southern Australia. J. Biogeogr. 11, 63–78.
Konvicka, M., Benes, J., Cizek, O., Kopecek, F., Konvicka, O., Vitaz, L., 2008. How too much care kills species: grassland reserves, agrienvironmental schemes and extinction of *Colias myrmidone* (Lepidoptera: Pieridae) from its former stronghold. J. Insect Conserv. 12, 519–525.
Kupcak, V., 2005. Ochrana lesa a lesní zákon. In: Kupcak, V. (Ed.), Ekonomické aspekty ochrany lesa, pp. 45–52.
Kupcak, V., 2010. Hospodářská opatření a některé ekonomické aspekty u lesa nízkého. In: Sloup, R., Sisak, L. (Eds.), Efektivnost lesního hospodářství se zřetelem k tvaru lesa nízkého, pp. 25–30.
Lindhe, A., Lindelow, A., Asenblad, N., 2005. Saproxylic beetles in standing dead wood density in relation to substrate sun-exposure and diameter. Biodiv. Conserv. 14, 3033–3053.
Lozek, V., 1973. Příroda ve čtvrtohorách. Academia, Czech Republic.
Macek, J., Straka, J., Bogusch, P., Dvorak, L., Bezdecka, P., Tyrner, P., 2010. Blanokřídlí České Republiky. Academia, Czech Republic.
Mielikainen, K., Hynynen, J., 2003. Silvicultural management in maintaining biodiversity and resistance of forests in Europe–boreal zone: case Finland. J. Environ. Manage. 67, 47–54.
Moga, C.I., Hartel, T., Öllerer, K., 2009. Ancient oak wood-pasture as a habitat for the endangered tree pipit *Anthus trivialis*. Biologia 64, 1011–1015.
Nemec, J., Lozek, V., 1996. Chráněná území ČR 1 – Střední Čechy. AOPK ČR, Praha.
Neuhauslova, Z. (Ed.), 1998. Map of Potential Natural Vegetation of the Czech Republic. Academia, Praha.
Nieto, A., Alexander, K., 2010. European Red List of Saproxylic Beetles. Publication Office of the EU, Luxembourg.
Norden, B., Götmark, F., Ryberg, M., Paltto, H., Allmér, J., 2008. Partial cutting reduces species richness of fungi on woody debris in oak-rich forests. Can. J. For. Res. 38, 1807–1816.
Pecha, M., 2010. Křivoklátské pařeziny a lesy sdružené. In: Sloup, R., Šišák, L. (Eds.), Efektivnost lesního hospodářství se zřetelem k tvaru lesa nízkého, pp. 12–15.
Peterken, G.F., 1993. Woodland Conservation and Management, second ed. Chapman and Hall, London.
Petrakova, L., Schlaghamersky, J., 2011. Interactions between *Liometopum microcephalum* (Formicidae) and other dominant ant species of sympatric occurrence. Commun. Ecol. 12, 9–17.
Ponder, W.F., Carter, G.A., Flemons, P., Chapman, R.R., 2001. Evaluation of museum collection data for use in biodiversity assessment. Conserv. Biol. 15, 648–657.
Ranius, T., Jansson, N., 2000. The influence of forest regrowth, original canopy cover and tree size on saproxylic beetles associated with old oaks. Biol. Conserv. 95, 85–94.
Rolecek, J., 2005. Vegetation types of dry-mesic oak forests in Slovakia. Preslia 77, 241–261.
Salek, L., Zahradnik, D., Marusak, R., Jerabkova, L., Merganic, J., 2013. Forest edges in managed riparian forests in the eastern part of the Czech Republic. For. Ecol. Manage. 305, 1–10.
Schmidl, J., Bußler, H., 2004. Ökologische Gilden xylobionter Käfer Deutschlands. Naturschutz. Landschaftsplan. 36, 202–218.
Slama, E.F., 1998. Cerambycidae of the Czech and Slovak Republics (Beetles-Coleoptera). Milan Sláma, Krhanice.
Svoboda, D., Peksa, O., Vesela, J., 2010. Epiphytic lichen diversity in central European oak forests: assessment of the effects of natural environmental factors and human influences. Environ. Pollut. 158, 812–819.
Svoboda, D., Peksa, O., Vesela, J., 2011. The analysis of epiphytic lichen composition in central European oak forests. Preslia 83, 129–144.
Svoboda, P., 1943. Křivoklátské lesy: dějiny jejich dřevin a porostů. Sumptibus societatis, Kruh mladých českých botaniků.
Ugurlu, E., Rolecek, J., Bergmeier, E., 2012. Oak woodland vegetation of Turkey – a first overview based on multivariate statistics. Appl. Veg. Sci. 15, 590–608.
Utinek, D., 2004. Conversions of coppices to a coppice-with-standards in urban forests of Moravský Krumlov. J. For. Sci. 50, 38–46.
Vera, F.W.M., 2000. Grazing Ecology and Forest History. CABI.
Vitousek, P.M., Mooney, H.A., Lubchenco, J., Melillo, J.M., 1997. Human domination of Earth's ecosystems. Science 277, 494–499.
Wilkie, D.S., Carpenter, J., 1999. The potential role of safari hunting as a source of revenue for protected areas in the Congo Basin. Oryx 33, 340–345.
Winter, S., Moller, G.C., 2008. Microhabitats in lowland beech forest as monitoring tool for nature conservation. For. Ecol. Manage. 255, 1251–1261.